Journal of the American Medical Informatics Association, 2025, **32(5)**, 932–944
https://doi.org/10.1093/jamia/ocaf017
Advance access publication 12 February 2025
Review

Review

Digital health equity frameworks and key concepts: a scoping review

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Abstract

Objectives: Digital health equity, the opportunity for all to engage with digital health tools to support good health outcomes, is an emerging priority across the world. The field of digital health equity would benefit from a comprehensive and systematic understanding of digital health, digital equity, and health equity, with a focus on real-world applications. We conducted a scoping review to identify and describe published frameworks and concepts relevant to digital health equity interventions.

Materials and Methods: We conducted a scoping review of published peer-reviewed literature guided by the PRISMA Extension for Scoping Reviews. We searched 5 databases for frameworks related to or applied to digital health or equity interventions. Using deductive and inductive approaches, we analyzed frameworks and concepts based on the socio-ecological model.

Results: Of the 910 publications initially identified, we included 44 (4.8%) publications in our review that described 42 frameworks that sought to explain the ecosystem of digital and/or health equity, but none were comprehensive. From the frameworks we identified 243 concepts grouped into 43 categories including characteristics of individuals, communities, and organizations; societal context; perceived value of the intervention by and impacts on individuals, community members, and the organization; partnerships; and access to digital health services, in-person services, digital services, and data and information, among others.

Discussion: We suggest a consolidated definition of digital health equity, highlight illustrative frameworks, and suggest concepts that may be needed to enhance digital health equity intervention development and evaluation.

Conclusion: The expanded understanding of frameworks and relevant concepts resulting from this study may inform communities and stakeholders who seek to achieve digital inclusion and digital health equity.

Key words: digital equity; health equity; consumer health informatics; community engagement; socioecological model.

Background

Digital health equity, the opportunity for all to engage with digital health tools to support good health outcomes,[1] is an emerging global priority.[2–4] Those working to develop, deploy, and attain adoption of real-world digital health equity interventions would benefit from a comprehensive and systematic understanding of digital health equity. Digital health equity exists within a complex socio-ecological system in which individuals are nested within interpersonal relationships, the community, and society—therefore, their engagement with digital solutions can be impacted by biases, whether explicit, implicit, or structural, integrated into policies, healthcare practices, and socio-cultural norms,[5–12] and impact an individual intersectionally.[13,14] However, consensus on its definition is currently lacking and digital health equity research is often critiqued for simplistic conceptualization of patient technology engagement.[13,14] This signals a need to develop unified conceptualizations of digital health equity that capture its complexity, given healthcare is increasingly reliant on digital solutions such as telehealth, remote patient monitoring, patient portals, virtual health education, and care coordination.[6]

Definitions

While there is no widely accepted definition of digital health equity, there are related terms that may provide a foundation for such a definition. First, *digital health*, defined by the World Health Organization (WHO) as "the field of knowledge and practice associated with the development and use of digital technologies to improve health" (p. 11).[15] The United States (U.S.) Food and Drug Administration's definition of digital health technologies encompasses "mobile health (mHealth), health information technology (IT), wearable

Received: August 16, 2024; **Revised:** December 30, 2024; **Editorial Decision:** January 18, 2025; **Accepted:** January 20, 2025

devices, telehealth and telemedicine, and personalized medicine."[11]

Second, *digital equity* is "the condition in which individuals and communities have the information technology capacity that is needed for full participation in the society and economy."[16] Digital equity impacts lives broadly, in ways that include supporting education and employment. Gaps in digital equity may arise from discrimination embedded in: (1) social norms (eg, racism, ablism) that can impact design of technologies and opportunities to learn about or engage with technology;[17–20] and (2) historic and contemporary policies (eg, Civil Rights Act of 1964) that can impact access to resources and opportunities among marginalized populations.[21–23] An example of how discrimination can impact digital equity is via digital gatekeepers such as internet companies engage in digital redlining within neighborhoods with residents who were from Black and other communities of color; many of these Black neighborhoods had formed due to past policies allowing racial segregation in housing among Black community members (redlining) that was made illegal in 1968 yet persists today.[24,25]

Third, *health equity* is "the attainment of the highest level of health for all people" that "requires valuing everyone equally with focused and ongoing societal efforts to address avoidable inequalities, historical and contemporary injustices, and the elimination of health and health care disparities."[26] Health equity impacts an individual's health and access to health services.[27] Gaps in health equity can involve discrimination embedded in contextually complex societal, economic, and institutional factors, including policy.[27,28]

Frameworks

Frameworks, conceptual models, and theories (hereafter referred to collectively as "frameworks") provide a common understanding and language about complex phenomenon and a structured way to develop and/or evaluate interventions.[29–32] For example, the socio-ecological model (SEM) was developed within the field of psychology and describes 4 nested levels of influence that impact each other: microsystem/individual, mesosystem/interrelations, exosystem/social structures, and macrosystem/institutional patterns of culture.[12] A number of researchers and public health agencies have applied this seminal model to understand the complexities of digital and health equity.[21,33–35] An updated SEM that includes an institutional/organizational level[36] has also been applied to public health and social institutions[37,38] and technology.[39–41]

Need for a unified definition and framework

There is a need for a clear conceptualization of digital health equity. The field is still developing as the rapid deployment of digital health interventions occurs within changing, complex ecosystems in which users live. Whether a user can access digital health technologies can be impacted by social determinants, or drivers, of health (SDOH)[4,42] (in this paper we use "driver" to align with efforts to assess and address SDOH in a way that is more immediately understandable and actionable).[43]

The WHO deems access to broadband internet as an SDOH necessary for full participation in society and the economy,[44] yet 12% of United States adults are without home internet access,[45] which may prevent them from benefiting from digital health technologies. Understanding the underlying causes for lack of internet is key to identifying appropriate interventions to improve access. In the United States, reasons may include digital redlining (the "intentional lack of investment in broadband infrastructure and affordable service," p. 2) among minority and low-income neighborhoods,[21] policing of free internet resources in non-minoritized and wealthier areas,[46] and the legacy of colonization of and discrimination against minoritized populations as codified and institutionalized in laws and their enforcement, education, and access to resources.[47] These practices can lead to decreased trust in digital resource providers, including government institutions and programs.[47–49] For example, the United States Affordable Connectivity Program was an intervention that provided subsidies for affordable internet subscriptions and a one-time discount for a device that connects to the internet. However, not all who qualified for the program could benefit due to lack of trust in federal programs or challenges enrolling among those whose primary language is not English or those with disabilities.[50]

Literature suggests that a lack of clear conceptualizations of digital health equity can have real-world impacts. During the COVID-19 public health emergency in 2020, healthcare organizations rapidly deployed telehealth.[51] However, marginalized populations experienced unequal access to telehealth and its benefits.[52–54] Researchers suggest that a possible factor is the lack of a strong theoretical foundation for digital health equity interventions.[13]

Therefore, to address this gap, we conducted a scoping review of the literature.[55] Our objectives were to identify and describe published frameworks relevant to digital health equity interventions, identify and describe definitions of "digital health equity," and understand the breadth and potential gaps in concepts represented in those frameworks.

Methods

This work is part of a larger study to develop guidance for community-based organizations in their digital health equity intervention work.[50]

Study design

This scoping review, guided by the PRISMA Extension for Scoping Reviews (PRISMA-ScR),[56] was conducted from March 2023 to March 2024. The search included all literature published prior to January 1, 2024, using broad search terms: (framework OR conceptual model OR theory) AND (equit* OR inequit* OR disparit*) AND (connectivity OR broadband OR internet OR technolog* OR informatics OR digital) AND (program evaluation OR planning techniques). We conducted an additional search for papers using the phrase "digital health equity" to capture the variety of disciplines engaging in digital health equity research. We queried the following databases: CINAHL (nursing), Engineering Village (computer science and engineering), PsycINFO (psychology), PubMed (biomedical), and Web of Science (social health sciences). Database yields were uploaded to Covidence (Melbourne, Australia).

Researchers with expertise in digital health and health equity and experienced in conducting scoping and systematic reviews conducted the screening in 2 phases. In phase 1, 1 reviewer screened abstracts, applying inclusion criteria to select articles considered for full-text review. In phase 2, 2 independent reviewers screened full text papers, applying inclusion criteria and documenting reasons for exclusion. The reviewers discussed discrepancies and came to consensus on decisions with the help of a third reviewer.

We included peer-reviewed articles that were written in English and published before January 1, 2024 that described studies that were: empirical; identified an existing or proposed a new conceptual framework/model/theory; involved technology, health, and equity; and described planning, implementation, and/or evaluation of a related program or intervention. Inclusion criteria were applied only to the studies and not to the frameworks; therefore, identified frameworks did not need to include digital health equity concepts if the study they were used in met the inclusion criteria. Systematic or scoping literature reviews and papers presenting only computational or mathematical models were excluded.

For the criterion of being a framework/model/theory, we accepted author-identified labeling and did not attempt to clearly differentiate between the 3 types given that authors often use the terms interchangeably.[57] We used the following definitions to determine whether the criterion was met: conceptual framework are a graphical or narrative explanation of "the main things to be studied—the key factors, concepts, or variables—and the presumed relationships among them;"[58] a conceptual model is a "diagram of proposed and causal linkages among a set of concepts believed to be related" to a specific issue under investigation and represents "empirical findings or the experience of practicing professionals;"[57] a theory is a "set of interrelated constructs (concepts), definitions, and propositions that present a systematic view of phenomena by specifying relations among variables, with the purpose of explaining and predicting the phenomena" (p.11).[32] We collectively describe these conceptualizations as "frameworks" in this paper.

A researcher re-read the final yield and extracted the following information: publication and study information; the name and purpose of the framework; evidence used, if any, to develop the framework; whether the framework was validated; names and definitions for framework concepts; and definitions and descriptions of "digital health equity" anywhere in the publication. We defined a concept as "a word or phrase that summarizes ideas, observations, and experiences" (p. 4). Propositions were defined as the relationships between concepts.[59]

Analysis

To address the study purposes, we conducted the following analytic steps.

Step 1. Characteristics of Published Frameworks. We labelled each framework identified in the papers as 1 of 3 types: "digital health equity-major" if digital health equity was explicitly described as a purpose of the framework (not to be confused with the purpose of the paper) and if named concepts related to digital equity, health equity, and/or digital health equity were present; "digital health equity-minor" if digital health equity was not an explicit purpose of the framework, but there were named concepts related to the same 3 terms present within the framework; and "general focus" if digital health equity was not included within the purpose of the framework or its concepts. We calculated basic counts and percents to quantify study and framework characteristics.

Step 2. Proposed Definition. We used the content analysis method to inductively identify common features across the definitions of digital health equity or their component words.[60,61]

Step 3. Concepts by SEM Level. We applied a deductive approach, the directed content analysis method, which uses existing theory to guide analysis categories, to organize concepts. We chose this method because applying a widely used framework that takes into account the complex context of digital health equity allowed us to assess gaps in the corpus of concepts used in digital health equity intervention research. We chose the version of the SEM that includes the organizational level in addition to the individual, interpersonal, community, and society levels.[36,58] One investigator independently assigned each concept using the definition provided by the source author to an SEM level. The other investigator independently reviewed the assignments, then both met to come to consensus of the finalized assignments.

Step 4. Breadth of Concepts. We conducted an inductive analysis using conventional content analysis,[62] during which concept categorization is guided by what emerges from the data. We chose this method to understand the breadth of concepts represented within this relatively new field of study without constraining it to an existing framework. To complete this analysis, we independently grouped similar concepts into high-level categories then jointly came to consensus on groupings and category labels.

Results

Yield

Figure 1 provides the PRISMA Flow Diagram. We identified 638 references after removing duplicates. Upon completion of all screening steps, 44 publications (30.1% of full-text reviewed papers, 4.8% of the total screened sample) were included in this analysis. A complete bibliography is available in File S1.

Study and framework characteristics

Most studies were conducted in North America ($n = 32$) and published in the past 5 years (see Table 1). Across the 44 studies, we identified 42 frameworks. 22 were existing frameworks and 20 were proposed frameworks. Thirty-four included definitions of some or all concepts. Among proposed new frameworks, most were informed by a combination of sources, including existing frameworks ($n = 11$), literature reviews ($n = 8$), and previous research ($n = 7$) (see Table 1).

Among the 42 frameworks identified, 13 had digital health equity as a major focus (digital health equity-major) (see Table 1). These included 7 new conceptualizations and 1 which built on an existing framework. Two papers noted that the frameworks were validated.[63,64] Only 8 major-focus frameworks defined concepts within their frameworks and were included in the concept analysis (see Table 2).

Eight frameworks had digital health equity as a minor component, focusing instead on general technology development/integration or assessment/evaluation of health technologies (Table 1). Seven of these digital health equity-minor frameworks provided concept definitions and were included in the concept analysis; only one was not[65] (Table 2). The remaining 19 frameworks included general-focus frameworks without a specific focus on digital heath equity, and all provided at least some concept definitions.

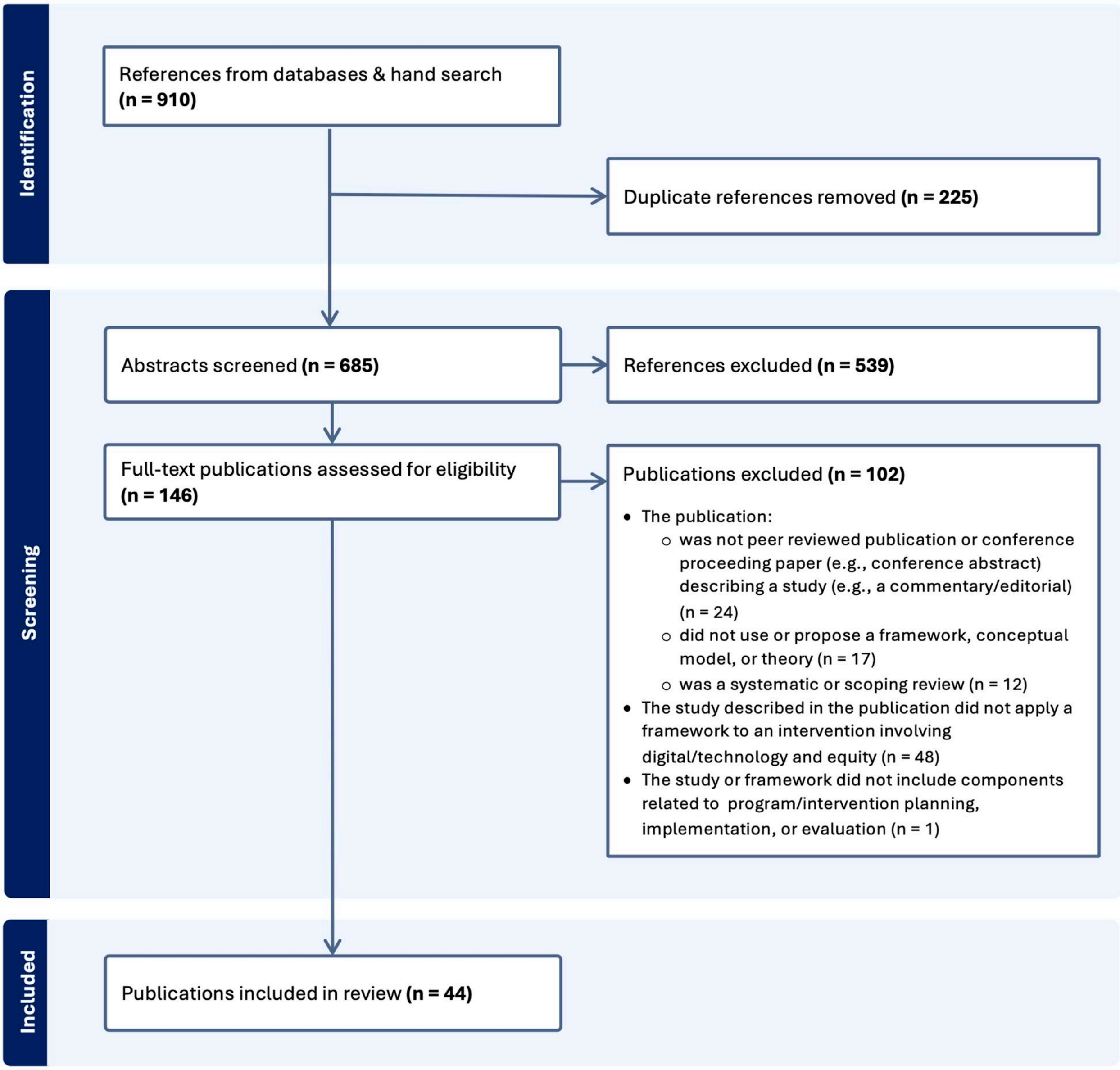

Figure 1. PRISMA flow diagram.

Definitions of "digital health equity"

Among the 44 publications included in our review, only 5 provided definitions or descriptions for digital health equity. Blanc et al. defines digital health equity and inclusion as "the fair and just opportunity to engage with digital health tools to support good health outcomes." (p. 262).[1] Foley et al.[66] cites a previous definition by Kickbusch, "Digital health equity is concerned with fair and just access to, use of and benefit from digital health services and is a critical axis of contemporary health promotion."[67] Ha et al. describes it as "the readiness of all individuals to access digital health, regardless of age, race, income, or technology access, ensuring that no one is left behind due to a lack of connectivity or literacy" (p. 2),[68] citing a previously published definition by Wood et al.[69] Kaihlanen et al. refers to it as "…an equal opportunity for individuals to benefit from the knowledge and practices related to the development and use of digital technologies to improve health" (p. 2),[70] citing previous definitions.[3,15] The fifth definition is suggested by Richardson et al.: "… [digital health equity includes] equitable access to digital healthcare, equitable outcomes from and experience with digital healthcare, and equity in the design of digital health solutions" (p. 2),[4] combining concepts from previous definitions.[2,3]

There are 5 common features across these definitions including: equity, a need or precursor to action, an action taken, a technology related solution, and a result of the action. Figure 2 shows these features with illustrative words from the definitions.

Concepts by SEM level

Of the 42 frameworks, 34 (81.0%) had at least some concepts defined. These frameworks also varied in the number of SEM levels covered by the framework (Table 2). About 44% of the frameworks ($n = 15$) had concepts addressing only 1 level of the SEM, while the other 56% ($n = 19$) were multilevel. Only 1 framework, National Quality Forum Telehealth

Table 1. Description of publications included in the scoping review ($n = 44$) and frameworks included in the publications ($n = 42$).

Characteristics	Count (%)
Among all publications ($n = 44$)	
Location of study or framework development[a]	
Africa	1 (2.3)
Asia	4 (9.1)
Australia	3 (6.8)
Caribbean	1 (2.3)
Europe	3 (6.8)
North America	32 (72.7)
South and Central America	1 (2.3)
Other	1 (2.3)
Year of publication	
2005, 2011, 2012, 2017	1 each
2018	2
2019	3
2020	3
2021	6
2022	13
2023	13
Among all frameworks ($n = 42$)	
Framework genesis	
Existing framework	22 (52.4)
Newly proposed framework	20 (47.6)
Framework's level of focus on digital health equity	
Major focus (digital health equity-major)	13 (31.0)
Concept definitions provided	8 (19.0)
Minor focus (digital health equity-minor)	8 (19.0)
Concept definitions provided	7 (16.7)
General focus	21 (50.0)
Concept definitions provided	19 (45.2)
Among proposed new frameworks ($n = 20$)	
Sources of information used to develop the framework[b]	
Narrative literature review	2 (10.0)
Literature review (type not specified)	6 (30.0)
Previous research by publication authors	7 (35.0)
Previously developed frameworks	11 (55.0)
Interviews	2 (10.0)
Focus groups	2 (10.0)
Survey	1 (5.0)
Other	8 (40.0)

[a] Some studies occurred in more than 1 continent, therefore the total count for this variable is greater than the number of publications.
[b] Some frameworks were developed using multiple types of evidence, therefore the total count for variable is greater than the number of publications that proposed a new framework.

Measurement Framework, referenced all SEM levels.[71] The most commonly represented level was the organizational level, with 27 (79.4%) frameworks covering it, and the least represented was interpersonal level ($n = 6$, 17.6%).

Breadth of concepts

Among the 34 frameworks with definitions, 243 concepts (see File S1 for complete list of concepts and mapping to categories) were identified and grouped into 43 categories (see Table 3). No single framework referenced all 43 categories. The most comprehensive frameworks were the digital health equity frameworks by Lyles et al.[72] and Richardson et al.[4] which addressed 15 and 12 concept categories, respectively. Several categories were only incorporated into DHE-major frameworks: access to digital services, digital health services, digital equity, implicit algorithmic and technology bias, digital health/e-Health literacy, perceived value of an intervention to the community/population/society, and technology policy. Fourteen concept categories were not included in any DHE-major frameworks, but rather, were only included in DHE-minor or general frameworks. Examples include concepts related to an intervention (eg, implementation process for and impacts of the intervention), communication among intervention stakeholders, and motivational factors for developing an intervention (eg, needs of the community or population). Several concepts were only part of general-focus frameworks, such as: needs of the implementing organization/program, perceived value of the intervention to the individual, and transparency.

Discussion

The key findings reported here are relevant to researchers or program developers who may find the breadth of applicable frameworks helpful to their design and planning of comprehensive and effective interventions.

In this scoping review, frameworks relevant to digital health equity including some from the fields of biomedical, population health, business, implementation science, information, and computer sciences. The variety of frameworks indicates that authors are generating knowledge from cross-disciplinary perspectives. In addition, the frameworks vary in the level of focus within the SEM and the number of levels addressed (ie, individual, interpersonal, organizational, community, and societal). The frameworks refer to a multi-level constellation of concepts: 243 concepts grouped into 43 categories. These concepts also span all 5 SEM levels. The individual and organizational/program levels were well-represented among the frameworks, while the interpersonal, community, and societal levels were less represented.

Previous literature highlights the importance of accounting for community and societal context in addressing health equity and digital equity,[99] aligning with recommendations that broadband interventions include educational outreach internally or via trusted community partners.[100] Our findings also align with the elements of successful state programs for digital equity as described by the U.S. National Telecommunications and Information Administration (NTIA) which coordinates the national Internet for All program.[101] One key aspects of Internet for All was the Affordable Connectivity Program (ACP), which addressed the individual-level concept that broadband and digital inclusion is an SDOH by providing internet subsidies for low-income families across the United States.[102]

Organizational/program level concept categories were represented in most papers: internal capacity and partnerships were important considerations when developing digital health equity interventions. Delivery of these interventions relies on the alignment of capabilities (eg, assets, skills, knowledge) and capacity (eg, funding, resources, time) of the community members, CBO, and partners to meet the community's needs. Building comprehensive approaches using a community coalition, as advocated by National Digital Inclusion Alliance (NDIA),[103] requires identifying and addressing mechanisms of minoritization, which can impact coalition relationships, and intervention design and efficacy.[104]

Fewer frameworks addressed interpersonal, community, and societal level concept categories. Several papers suggested that understanding community needs is key to intervention planning and evaluation. Yet, the frameworks infrequently defined community-level concepts explicitly. While publicly-

Table 2. Concepts organized by socio-ecological level among frameworks ($n = 38$).

Sociological level of influence (Papers with at least 1 concept defined and in the framework at the level)		Individual	Interpersonal	Org. or program	Community	Societal
Author	**Framework**					
Digital health major frameworks ($n = 8$)						
Blanc et al. 2023[1]	Digital health equity and inclusion framework/model	X	X	X		
Faber et al. 2023[73]	Guide for ehealth intervention design			X		
Foley et al. 2021[66]	Suggested pathways of access, use, and benefit from digital health services	X	X	X		
Henson et al. 2023[74]	Djurali Model for digital health equity	X		X	X	
Lyles et al. 2022[72]	Framework for digital health equity	X		X	X	X
Richardson et al. 2022[4]	Framework for digital health equity*	X	X		X	X
Pacifico Silva et al. 2018[63]	Responsible innovation in health (RIH) framework	X		X		X
Sun et al. 2023[64]	The impact of digital inclusion on health status: the mediating effect of cultural capital	X				
Digital health minor frameworks ($n = 7$)						
De Souza et al. 2021[75]	Techno-economic framework for installing broadband networks in rural and remote areas				X	
Gauvreau et al. 2023[76]	Child-tailored health technology assessment (HTA) framework	X	X	X	X	
Kukhareva et al. 2022[77]	Evaluation in life cycle of IT (ELICIT) framework			X		
Nair et al. 2023[78]	ConNECT framework			X	X	
Steinman et al. 2023[79]	Reach-effectiveness-adoption-implementation-maintenance + equity framework*			X	X	
Whitehead et al. 2022[71]	National quality forum Telehealth measurement framework*	X	X	X	X	X
Zhang and Liu 2022[80]	Smart city ontology				X	
General-focus frameworks ($n = 19$)						
Allicock et al. 2017[81]	The diffusion of innovation theory*			X	X	
Spierling Bagsic et al. 2023[82]	Reach-effectiveness-adoption-implementation-maintenance framework*			X		
Belkin et al. 2011[83]	5×5 framework	X		X		
Braganza et al. 2022[84]	Learning health system framework*	X		X		
Chanfreau-Coffinier et al. 2019[85]	A logic model for precision medicine implementation			X		
Cheng et al. 2021[86]	Self-determination theory*	X				
Eiraldi et al. 2022[87]	Interactive system framework for dissemination and implementation (ISF)*			X		
Gallo et al. 2021[88]	Four feedback mechanisms involving inputs, processes, outputs, and outcomes			X		
Kim et al. 2023[89]	Medical research council (MCR) process evaluation framework*			X	X	
Kohn et al. 2023[90] Steinman et al. 2023[79]	Framework for reporting adaptations and modifications-expanded (FRAME)	X		X		
Kohn et al. 2023[90] Steinman et al 2023[79]	Implementation outcomes framework	X		X		
Laur et al. 2022[91]	Quadruple aim framework	X		X		X
Nelson et al. 2022[92]	Consolidated framework for implementation research (CFIR)*	X		X	X	
Nelson and Zanti 2020[93]	Government alliance for race equity framework*					X
Rhodes et al. 2021[94]	Community energy balance (CEB) framework*	X		X	X	X
Ross et al. 2018[95]	Generalizable framework for multi-scale auditing of digital learning provision higher Ed.			X		
Stone and Lan 2012[96]	Planning and evaluating technology-based research and development*			X		
Tebb et al. 2019[97]	6 PCORI (patient-centered outcomes research institute) Engagement Principles*		X			
Wieland et al. 2022[98]	Crisis and emergency risk communication (CERC) framework*			X		
*Existing framework	Number of frameworks covering each socio-ecological level	**17**	**6**	**27**	**13**	**7**

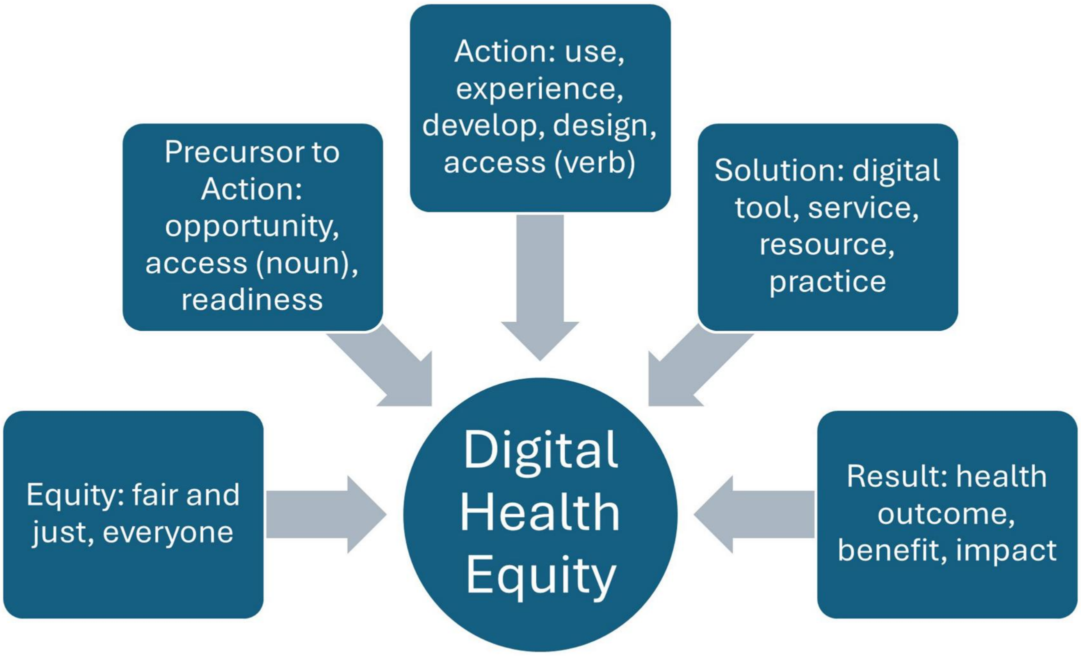

Figure 2. Common concepts in definitions of digital health equity with illustrative words.

available national data quantifying community needs and internet availability exist,[105] these concepts may not be adequate for characterizing specific local needs. For example, in prior qualitative research, we identified gaps in data for identifying the particular needs of local communities and a lack of valid and reliable measures of digital health equity; key informants reported having to create customized surveys to try and capture these data, resulting in lack of comparability.[47] While several assessment guides exist, including one from the NTIA106 and the National League of Cities Digital Equity Playbook,[107] there is a need for validated measures that CBOs can easily use to plan and evaluate their digital health equity interventions.

Similarly, several papers mention the importance of relationships and trust between CBOs and communities, but do not specify concepts related to these interpersonal relationships. For example, papers suggest that community members must trust digital equity interventions, federal programs, internet service providers, and other providers of digital health, including clinicians.[108] This finding aligns with previous work, highlighting the importance of trust in internet service providers and digital tools and policies supportive of consumer protection.[48,104] Interestingly, the concepts we identified did not include trust related to data breaches. This aligns with our previous research which found that trust around data privacy and security was focused around the use of data for oppression. Understanding the sources of mistrust is important, as it identifies the upstream causes for mistrust, such as mistrust of the government or medical facilities among Black, Indigenous, and other communities of color due to legacies of colonization and slavery, and institutionalized racism integrated into medical practice.[109] Additionally, there has been recent attention paid to the role of bias in technology and algorithms that may exacerbate health disparities.[110–112] It may be interesting to understand how trustworthy interpersonal relationships are characterized and manifested through communication and behavior across stakeholders, and whether the influence of these relationships can ameliorate such biases.

Finally, the societal level is not often included in frameworks. Only 3 DHE-major [4,63,72], 1 DHE-minor,[71] and 3 general frameworks [91,93,94] include concepts at the societal level, such as the societal context, technology policy, and impacts of an intervention on society. There are several societal factors and characteristics that can affect the individual, organizational, community, and interpersonal levels of the ecosystem. For example, legal and regulatory policies governing healthcare and technology, market dynamics that affect the supply and demand of digital health, and population characteristics would certainly affect digital health interventions.

Gaps

As one of the first scoping reviews to address frameworks and concepts of digital health equity, we noted several gaps based on the principal findings of our study. First, there is no consensus or widely accepted definition of digital health equity; there was variation across studies in how they defined the concept, if they defined it at all.

Second, there were concept categories present in digital health equity-minor and general focus frameworks but missing from digital health equity-major frameworks as shown in Table 3. Several of these categories related to the organization and program (eg, needs of the organization implementing the program, implementation process for the program), suggesting that health equity-major frameworks may be more conceptual than actionable for those looking to use a framework to guide an intervention.

Third, there is no comprehensive framework that encompasses all 5 SEM levels. In concert, these indicate an overall gap in understanding about how the intersection and application of digital health and equity concepts can inform the

Table 3. Digital health equity concept categories by focus of frameworks.

	Concept category	Citation, organized by digital health equity-major, -minor, and general-focus categories
1	Access to digital health services (eg, telehealth)	Major: Foley 2021[66] Minor: Whitehead 2022[71]
2	Access to digital services (eg, broadband internet)	Major: Henson 2023[74], Lyles 2022[72], Richardson 2022[4], Sun 2023[64]
3	Access to health information and data[a]	Minor: Whitehead 2022[71] General: Braganza 2022[84]
4	Access to in-person health care	Major: Foley 2021[66] Minor: Whitehead 2022[71]
5	Autonomy of individuals[b]	General: Cheng 2021[86]
6	Characteristics of individuals involved in intervention (eg, patients, staff)	Major: Blanc 2023[1], Foley 2021[66], Lyles 2022[72], Richardson 2022[4], Sun 2023[64] Minor: Gauvreau 2023[76], Zhang 2022[80] General: Braganza 2022[84], Nelson 2022[92], Rhodes 2021[94]
7	Characteristics of interpersonal relationships	Major: Blanc 2023[1], Richardson 2022 Minor: Gauvreau 2023[76] General: Rhodes 2021[94]
8	Characteristics of community	Major: Blanc 2023[1], Foley 2021[66], Lyles 2022[72], Richardson 2022[4], Sun 2023[64] General: Kohn 2023[90], Rhodes 2021[94]
9	Characteristics of population[a]	Minor: Gauvreau 2023[76], Zhang 2022[80] General: Laur 2022[91], Rhodes 2021[94]
10	Characteristics of intervention or program	Major: Blanc 2023[1], Henson 2023[74], Pacifico Silva 2018[63] Minor: Nair 2023[78], Zhang 2022[80] General: Chanfreau-Coffinier 2019[85], Eiraldi 2022[87], Kim 2023[89], Kohn 2023[90], Laur 2022[91], Nelson 2022[92], Rhodes 2021[94], Stone 2012[96]
11	Characteristics of organization	Major: Blanc 2023[1] General: Gallo 2021[88], Kohn 2023[90], Nelson 2020[93], Nelson 2022[92], Stone 2012[96]
12	Co-learning among intervention stakeholders	Major: Lyles 2022[72] General: Braganza 2022[84], Tebb 2019[97]
13	Communication among intervention stakeholders[a]	Minor: Nair 2023[78] General: Allicock 2017[81], Wieland 2022[98]
14	Competence[b]	General: Cheng 2021[86]
15	Digital health services	Major: Foley 2021[66]
16	Emotion[b]	General: Wieland 2022[98]
17	Equity, digital	Major: Henson 2023[74], Lyles 2022[72], Richardson 2022[4], Sun 2023[64]
18	Equity, digital health	Major: Blanc 2023[1], Lyles 2022[72] Minor: Gauvreau 2023[76], Whitehead 2022[71]
19	Equity, health	Major: Blanc 2023[1], Lyles 2022[72], Pacifico Silva 2018[63] Minor: Gauvreau 2023[76], Nair 2023[78], Steinman 2023[79] General: Kohn 2023[90], Nelson 2020[93]
20	Event causing risk to community[b]	General: Wieland 2022[98]
21	Impacts of intervention[a]	Minor: Gauvreau 2023[76], Steinman 2023[79], Whitehead 2022[71] General: Spierling Bagsic 2023[82], Chanfreau-Coffinier, 2019[85], Kohn 2023[90]
22	Implementation process for intervention[a]	Minor: Nair 2023[78], Steinman 2023[79] General: Allicock 2017[81], Spierling Bagsic 2023[82], Chanfreau-Coffinier 2019[85], Eiraldi 2022[87], Gallo 2021[88], Kim 2023[89], Kohn 2023[90], Nelson 2020[93], Nelson 2022[92], Rhodes 2021[94], Stone 2012[96]
23	Implicit algorithmic bias	Major: Richardson 2022[4]
24	Implicit technology bias	Major: Blanc 2023[1], Richardson 2022[4]
25	Literacy, digital health/e-Health	Major: Foley 2021[66], Pacifico Silva 2018[63]
26	Literacy, digital	Major: Lyles 2022[72], Richardson 2022[4], Sun 2023[64] General: Ross 2018[95]
27	Needs of organization implementing program[b]	General: Kohn 2023[90], Stone 2012[96]
28	Needs of community or population[a]	Minor: Gauvreau 2023[76] General: Kohn 2023[90], Stone 2012[96]
29	Outcomes and effectiveness of intervention	Major: Blanc 2023[1], Henson 2023[74], Sun 2023[64] Minor: Gauvreau 2023[76], Steinman 2023[79], Whitehead 2022[71] General: Allicock 2017[81], Spierling Bagsic 2023[82], Chanfreau-Coffinier 2019[85], Gallo 2021[88] Kohn 2023[90], Kim 2023[89], Laur 2022[91], Stone 2012[96]
30	Partnerships	Major: Lyles 2022[72], Richardson 2022[4] General: Allicock 2017[81], Rhodes 2021[94], Tebb 2019[97]
31	Perceived value of intervention to individuals[b]	General: Spierling Bagsic 2023[82]
32	Perceived value of intervention to organization or program	Major: Pacifico Silva 2018[63] Minor: De Souza 2021[75], Whitehead 2022[71] General: Braganza 2022[84]

(continued)

Table 3. (continued)

	Concept category	Citation, organized by digital health equity-major, -minor, and general-focus categories
33	Perceived value of intervention to community or population	Major: Foley 2021[66], Pacifico Silva 2018[63]
34	Perceived value of intervention to society	Major: Pacifico Silva 2018[63]
35	Reciprocal relationships among stakeholders	Major: Lyles 2022[72], Richardson 2022[4] General: Allicock 2017[81], Tebb 2019[97]
36	Relatedness[b]	General: Cheng 2021[86]
37	Societal context	Major: Blanc 2023[1], Henson 2023[74], Lyles 2022[72], Pacifico Silva 2018[63], Richardson 2022[4] Minor: DeSouza 2021[75], Gauvreau 2023[76], Zhang 2022[80] General: Allicock 2017[81], Kim 2023[89], Kohn 2023[90], Nelson 2022[92], Stone 2012[96], Rhodes 2021[94]
38	Sustainability of intervention	Major: Pacifico Silva 2018[63] Minor: Steinman 2023[79], Whitehead 2022[71], Zhang 2022[80] General: Allicock 2017[81], Spierling Bagsic 2023[82], Gallo 2021[88], Kohn 2023[90]
39	Technology policy	Major: Lyles 2022[72], Richardson 2022[4]
40	Transparency[b]	General: Braganza 2022[84], Tebb 2019[97]
41	Trust	Major: Blanc 2023[1], Foley 2021[66], Lyles 2022[72] General: Tebb 2019[97]
42	User experience	Major: Lyles 2022[72] Minor: Whitehead 2022[71] General: Spierling Bagsic 2023[82], Braganza 2022[84], Laur 2022[91]
43	Workforce skills	Major: Lyles 2022[72] Minor: Nair 2023[78] General: Belkin 2011[83], Braganza 2022[84], Eiraldi 2022[87]

[a] Concept present only in digital health equity-minor or general-focus frameworks;
[b] Concept present only in general-focus frameworks.

development and implementation of equitable digital health interventions.

Consolidated definition of digital health equity

Based on the findings of this scoping review, we first suggest a consolidated definition of digital health equity that addresses the 5 concepts of equity, a need or precursor to action, an action taken, a technology related solution, and a result of the action. We used wording from the papers and other cited literature. Our proposed definition is: Digital health equity is a multi-level socio-ecological concept that results from fair and just opportunities for everyone to attain their highest level of health through access to technology-enabled health resources and services.[4,26,113–115] Digital health equity interventions apply technology-enabled solutions to deliver health equity impacts.

We discovered a number of intriguing concepts in this review that could contribute to a deeper characterization of digital health equity. Additional research is necessary to map the large number of identified concepts that could apply to digital health equity and validate which concepts are most relevant and useful. A more comprehensive framework including these concepts should highlight that digital health equity is not an effort that can be delivered by one sector alone but requires partnership across stakeholders from healthcare, economics, telecommunications and other industry players, CBOs, and community members themselves, among others. The findings we report may inform the development of a framework, and eventually an explanatory model of digital health equity intervention effectiveness. Additional investigation might determine the relevance of the 43 categories and 243 concepts we identified to see which are relevant and necessary for inclusion in a digital health equity framework.

Limitations

The purpose of a scoping review is to describe the landscape of knowledge in a field to identify concepts and highlight gaps. By design, this type of review is limited by searching for literature at one point in time. In a new topic, such as digital health equity, search terms are not standardized, so we may have missed papers that did not use the keywords we used in our search. To minimize these limitations, we cross-checked systematic reviews and reference lists of included paper, to be as thorough as possible.

Conclusions

Digital health equity is emerging as a priority across the world. As national and local investments are made to increase access to broadband for those who are currently underserved, we have the opportunity to leverage those investments to achieve improved health and health equity. As one of the first scoping reviews to address the need for frameworks for digital health equity, we offer several contributions to the field. First, we offer a consolidated definition of digital health equity. Second, the review highlights the diversity of frameworks that researchers might apply to their research questions and intervention developers might apply their program designs and evaluations. This greater awareness and understanding of frameworks may support improved study and program designs. Third, our research provides an inventory of the broad set of factors, represented by the concepts, concept categories and organization by SEM, that might impact whether and how digital health equity may be achieved. These data may form the basis of a more comprehensive framework for digital health equity. In future, these contributions may offer guidance to those communities and stakeholders who are working toward achieving both digital inclusion and digital health equity.

Acknowledgments

We thank Surya Mehta for assistance with data collection and Elizabeth Gloor for copy-editing support.

Author contributions

K.K. and U.B. are co-first authors and made equal and substantial contributions to the conceptualization, methodology, data acquisition, analysis, and investigation. Both K.K. and U.B. wrote major portions of the manuscript, had final approval of the version to be published, and are accountable for all aspects of the work.

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

Funding

This work was supported by The MITRE Innovation Program, an intramural funding program of the MITRE Corporation, which had no control over the conduct of the research nor the findings of the research.

Conflicts of interest

K.K. was previously employed by The MITRE Corporation, a non-profit research and development corporation, at the time the study was conducted. U.B. is currently employed by The MITRE Corporation. K.K. and U.B. received salary support from the MITRE Innovation Program during the study period.

Data availability

The original data generated in the course of the study is provided as Supplementary Material (File S1).

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Journal of the American Medical Informatics Association, 2025, 32, 932–944
https://doi.org/10.1093/jamia/ocaf017
Review